

TEN IMPORTANT SKILLS:
SAMPLE QUIZZES (KEY)

1. a) Let A be the following complex matrix and let v be the following vector in $\mathbb{C}^3$. Find Av .

$$A = \begin{pmatrix} 0 & i & 1 \\ 1+i & 1 & 0 \end{pmatrix}, \quad v = \begin{pmatrix} 1 \\ i \\ 1 \end{pmatrix}$$

$$\begin{aligned} Av &= \begin{pmatrix} 0 & i & 1 \\ 1+i & 1 & 0 \end{pmatrix} \begin{pmatrix} 1 \\ i \\ 1 \end{pmatrix} = 1 \begin{pmatrix} 0 \\ 1+i \end{pmatrix} + i \begin{pmatrix} i \\ 1 \end{pmatrix} + 1 \begin{pmatrix} 1 \\ 0 \end{pmatrix} \\ &= \begin{pmatrix} 0 \\ 1+i \end{pmatrix} + \begin{pmatrix} -1 \\ i \end{pmatrix} + \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ 1+2i \end{pmatrix} \end{aligned}$$

$$\text{or: } Av = \begin{pmatrix} 0 & i & 1 \\ 1+i & 1 & 0 \end{pmatrix} \begin{pmatrix} 1 \\ i \\ 1 \end{pmatrix} = \begin{pmatrix} 0 - 1 + 1 \\ 1+i + i + 0 \end{pmatrix} = \begin{pmatrix} 0 \\ 1+2i \end{pmatrix}$$

- b) Let B be the following real matrix:

$$B = \begin{pmatrix} 8 & 0 \\ 0 & 1 \\ 0 & 4 \end{pmatrix}$$

Complete the following sentence by filling in the four blanks:

The matrix B is a 3 $\times$ 2 matrix and it represents a linear transformation from $\mathbb{R}^{\underline{2}}$ to $\mathbb{R}^{\underline{3}}$.

2. a) For each of the following matrix operation, calculate the result if the operation is defined. Otherwise, state that the operation is undefined.

$$A = \begin{pmatrix} 2 & 0 \\ 0 & 1 \end{pmatrix} \quad B = \begin{pmatrix} 2 & 0 \\ 1 & 1 \end{pmatrix} \quad C = \begin{pmatrix} 2 & 0 & -6 \\ 0 & 1 & 2 \end{pmatrix} \quad D = \begin{pmatrix} 2 \\ 3 \end{pmatrix}$$

$B - A$, BA , $B + C$, AC , BD

$$B - A = \begin{pmatrix} 2 & 0 \\ 1 & 1 \end{pmatrix} - \begin{pmatrix} 2 & 0 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}$$

$$BA = \begin{pmatrix} 2 & 0 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} 2 & 0 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 4+0 & 0+0 \\ 2+0 & 0+1 \end{pmatrix} = \begin{pmatrix} 4 & 0 \\ 2 & 1 \end{pmatrix}$$

$B + C$ is undefined

$$AC = \begin{pmatrix} 2 & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 2 & 0 & -6 \\ 0 & 1 & 2 \end{pmatrix} = \begin{pmatrix} 4+0 & 0+0 & -12+0 \\ 0+0 & 0+1 & 0+2 \end{pmatrix} = \begin{pmatrix} 4 & 0 & -12 \\ 0 & 1 & 2 \end{pmatrix}$$

$$BD = \begin{pmatrix} 2 & 0 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} 2 \\ 3 \end{pmatrix} = \begin{pmatrix} 4+0 \\ 2+3 \end{pmatrix} = \begin{pmatrix} 4 \\ 5 \end{pmatrix}$$

b) Give an example of two nonzero 2×2 matrices whose product is a zero matrix, or explain why this is not possible.

One example is: $\begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 0+0 & 0+0 \\ 0+0 & 0+0 \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}$

3. a) Find the 2×2 matrix for the following linear transformation using the standard basis for $\mathbb{R}^2$.

* rotate about the origin by $\pi/6$ counterclockwise and then [#] reflect across the y-axis

$$\begin{pmatrix} 1 \\ 0 \end{pmatrix} \rightarrow \begin{pmatrix} \sqrt{3}/2 \\ 1/2 \end{pmatrix} \rightarrow \begin{pmatrix} -\sqrt{3}/2 \\ 1/2 \end{pmatrix}$$

$$\begin{pmatrix} 0 \\ 1 \end{pmatrix} \xrightarrow{*} \begin{pmatrix} -1/2 \\ \sqrt{3}/2 \end{pmatrix} \xrightarrow{\#} \begin{pmatrix} 1/2 \\ \sqrt{3}/2 \end{pmatrix}$$

$$\rightsquigarrow \begin{pmatrix} -\sqrt{3}/2 & 1/2 \\ 1/2 & \sqrt{3}/2 \end{pmatrix}$$

b) Describe in words the linear transformation given by the following matrix, which uses the standard basis for $\mathbb{R}^4$.

$$\begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

$e_1 \ e_2 \ e_3 \ e_4$

This linear transformation is projection onto the e_1 - e_3 plane, that is, the plane $\text{span}(\{e_1, e_3\})$.

4. a) Use row reduction to put the following matrix in reduced row echelon form.

$$\begin{pmatrix} 2 & 0 & -6 & -8 \\ 0 & 1 & 2 & 3 \\ 3 & 6 & -2 & -4 \end{pmatrix}$$

$$\begin{pmatrix} 2 & 0 & -6 & -8 \\ 0 & 1 & 2 & 3 \\ 3 & 6 & -2 & -4 \end{pmatrix} \xrightarrow{\frac{1}{2}R_1} \begin{pmatrix} 1 & 0 & -3 & -4 \\ 0 & 1 & 2 & 3 \\ 3 & 6 & -2 & -4 \end{pmatrix} \xrightarrow{R_3 - 3R_1} \begin{pmatrix} 1 & 0 & -3 & -4 \\ 0 & 1 & 2 & 3 \\ 0 & 6 & 7 & 8 \end{pmatrix}$$

$$\xrightarrow{R_3 - 6R_2} \begin{pmatrix} 1 & 0 & -3 & -4 \\ 0 & 1 & 2 & 3 \\ 0 & 0 & -5 & -10 \end{pmatrix} \xrightarrow{-\frac{1}{5}R_3} \begin{pmatrix} 1 & 0 & -3 & -4 \\ 0 & 1 & 2 & 3 \\ 0 & 0 & 1 & 2 \end{pmatrix} \xrightarrow{R_2 - 2R_3} \begin{pmatrix} 1 & 0 & -3 & -4 \\ 0 & 1 & 0 & -1 \\ 0 & 0 & 1 & 2 \end{pmatrix}$$

$$\xrightarrow{R_1 + 3R_3} \begin{pmatrix} 1 & 0 & 0 & 2 \\ 0 & 1 & 0 & -1 \\ 0 & 0 & 1 & 2 \end{pmatrix} \leftarrow \text{RREF}$$

b) Give the general solution to the following linear system in vector form.

$$x_1 + 2x_2 + 3x_3 = 4$$

$$-x_1 + x_2 + 2x_3 = 5$$

$$\left(\begin{array}{ccc|c} 1 & 2 & 3 & 4 \\ -1 & 1 & 2 & 5 \end{array} \right) \xrightarrow{R_2 + R_1} \left(\begin{array}{ccc|c} 1 & 2 & 3 & 4 \\ 0 & 3 & 5 & 9 \end{array} \right) \xrightarrow{\frac{1}{3}R_2} \left(\begin{array}{ccc|c} 1 & 2 & 3 & 4 \\ 0 & 1 & 5/3 & 3 \end{array} \right)$$

$$\xrightarrow{R_1 - 2R_2} \left(\begin{array}{ccc|c} 1 & 0 & -1/3 & -2 \\ 0 & 1 & 5/3 & 3 \end{array} \right)$$

$$x_1 = -2 + \frac{1}{3}x_3$$

$$x_2 = 3 - \frac{5}{3}x_3$$

x_3 is free

In vector form:
$$\mathbf{x} = \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} -2 + \frac{1}{3}x_3 \\ 3 - \frac{5}{3}x_3 \\ x_3 \end{pmatrix} = \begin{pmatrix} -2 \\ 3 \\ 0 \end{pmatrix} + x_3 \begin{pmatrix} 1/3 \\ -5/3 \\ 1 \end{pmatrix}$$

5. a) Compute the inverse of the following matrix.

$$\begin{pmatrix} 0 & 1 & 0 \\ 2 & 1 & 3 \\ -1 & 2 & 2 \end{pmatrix}$$

$$\left(\begin{array}{ccc|ccc} 0 & 1 & 0 & 1 & 0 & 0 \\ 2 & 1 & 3 & 0 & 1 & 0 \\ -1 & 2 & 2 & 0 & 0 & 1 \end{array} \right) \xrightarrow{R1 \leftrightarrow R2} \left(\begin{array}{ccc|ccc} 2 & 1 & 3 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 & 0 & 0 \\ -1 & 2 & 2 & 0 & 0 & 1 \end{array} \right) \xrightarrow{R1+R3} \left(\begin{array}{ccc|ccc} 1 & 3 & 5 & 0 & 1 & 1 \\ 0 & 1 & 0 & 1 & 0 & 0 \\ -1 & 2 & 2 & 0 & 0 & 1 \end{array} \right)$$

$$\xrightarrow{R3+R1} \left(\begin{array}{ccc|ccc} 1 & 3 & 5 & 0 & 1 & 1 \\ 0 & 1 & 0 & 1 & 0 & 0 \\ 0 & 5 & 7 & 0 & 1 & 2 \end{array} \right) \xrightarrow{R3-5R2} \left(\begin{array}{ccc|ccc} 1 & 3 & 5 & 0 & 1 & 1 \\ 0 & 1 & 0 & 1 & 0 & 0 \\ 0 & 0 & 7 & -5 & 1 & 2 \end{array} \right) \xrightarrow{1/7 R3} \left(\begin{array}{ccc|ccc} 1 & 3 & 5 & 0 & 1 & 1 \\ 0 & 1 & 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & -5/7 & 1/7 & 2/7 \end{array} \right)$$

$$\xrightarrow{R1-3R2} \left(\begin{array}{ccc|ccc} 1 & 0 & 5 & -3 & 1 & 1 \\ 0 & 1 & 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & -5/7 & 1/7 & 2/7 \end{array} \right) \xrightarrow{R1-5R3} \left(\begin{array}{ccc|ccc} 1 & 0 & 0 & 4/7 & 2/7 & -3/7 \\ 0 & 1 & 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & -5/7 & 1/7 & 2/7 \end{array} \right) \text{ so: } \begin{pmatrix} 0 & 1 & 0 \\ 2 & 1 & 3 \\ -1 & 2 & 2 \end{pmatrix}^{-1} = \begin{pmatrix} 4/7 & 2/7 & -3/7 \\ 1 & 0 & 0 \\ -5/7 & 1/7 & 2/7 \end{pmatrix}$$

b) Let A and B be the following matrices. Compute the inverse matrix of the product AB .

$$A = \begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix}, \quad B = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$$

① compute product, find inverse:

$$AB = \begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix} \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$$

$$\text{and so } (AB)^{-1} = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$$

② compute individual inverses, compute $B^{-1}A^{-1}$:

$$A^{-1} = \begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix}$$

$$B^{-1} = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$$

$$B^{-1}A^{-1} = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} \begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix} = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$$

6. a) Give a basis for the column space of A .

$$A = \begin{pmatrix} 6 & 2 & 2 & 0 \\ 1 & 3 & 11 & 8 \\ 1 & 4 & 15 & 11 \end{pmatrix}$$

row reduce A :

$$\begin{pmatrix} 6 & 2 & 2 & 0 \\ 1 & 3 & 11 & 8 \\ 1 & 4 & 15 & 11 \end{pmatrix} \xrightarrow{R1 \leftrightarrow R2} \begin{pmatrix} 1 & 3 & 11 & 8 \\ 6 & 2 & 2 & 0 \\ 1 & 4 & 15 & 11 \end{pmatrix} \xrightarrow{R2 - 6R1} \begin{pmatrix} 1 & 3 & 11 & 8 \\ 0 & -16 & -64 & -48 \\ 1 & 4 & 15 & 11 \end{pmatrix}$$

$$\begin{pmatrix} 1 & 3 & 11 & 8 \\ 0 & -16 & -64 & -48 \\ 1 & 4 & 15 & 11 \end{pmatrix} \xrightarrow{-\frac{1}{16}R2} \begin{pmatrix} 1 & 3 & 11 & 8 \\ 0 & 1 & 4 & 3 \\ 1 & 4 & 15 & 11 \end{pmatrix} \xrightarrow{R3 - R1} \begin{pmatrix} 1 & 3 & 11 & 8 \\ 0 & 1 & 4 & 3 \\ 0 & 1 & 4 & 3 \end{pmatrix}$$

$$\xrightarrow{R3 - R2} \begin{pmatrix} 1 & 3 & 11 & 8 \\ 0 & 1 & 4 & 3 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

Pivot columns of REF tell us the columns of A forming a basis for $\text{Col } A$

$\left\{ \begin{pmatrix} 6 \\ 1 \\ 1 \end{pmatrix}, \begin{pmatrix} 2 \\ 3 \\ 4 \end{pmatrix} \right\}$ is a basis for $\text{Col } A$.

b) Give an example of a basis for $\mathbb{R}^3$ containing the vectors $(1, 2, 3)^T$ and $(4, 5, 6)^T$ or explain why this is not possible.

$\begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}$ and $\begin{pmatrix} 4 \\ 5 \\ 6 \end{pmatrix}$ are linearly independent, since one is not a scalar multiple of the other. Since any three linearly independent vectors form a basis for $\mathbb{R}^3$, it is enough to produce a vector not in $\text{span}\left\{\begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}, \begin{pmatrix} 4 \\ 5 \\ 6 \end{pmatrix}\right\}$.

We show that the equation $a\begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix} + b\begin{pmatrix} 4 \\ 5 \\ 6 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$

has no solution: $\begin{pmatrix} 1 & 4 & 0 \\ 2 & 5 & 0 \\ 3 & 6 & 1 \end{pmatrix} \xrightarrow{R2 - 2R1} \begin{pmatrix} 1 & 4 & 0 \\ 0 & -3 & 0 \\ 3 & 6 & 1 \end{pmatrix} \xrightarrow{-\frac{1}{3}R2} \begin{pmatrix} 1 & 4 & 0 \\ 0 & 1 & 0 \\ 3 & 6 & 1 \end{pmatrix}$

$$\xrightarrow{R1 - 4R2} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 3 & 6 & 1 \end{pmatrix} \xrightarrow{R3 - 3R1} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 6 & 1 \end{pmatrix} \xrightarrow{R3 - 6R2} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

So: $\left\{ \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}, \begin{pmatrix} 4 \\ 5 \\ 6 \end{pmatrix}, \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} \right\}$ is a basis for $\mathbb{R}^3$.

—Midterm exam—

7. a) Using the standard basis, a linear transformation is given by the matrix A . Write the same linear transformation using v_1 and v_2 as an ordered basis for $\mathbb{R}^2$.

$$A = \begin{pmatrix} 0 & 1 \\ 2 & 1 \end{pmatrix}, \quad v_1 = \begin{pmatrix} 1 \\ 1 \end{pmatrix} \quad v_2 = \begin{pmatrix} 1 \\ 2 \end{pmatrix}$$

We have $Q = \begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix}$

and so the matrix for the transformation in the new basis is

$$Q^{-1} A Q = \begin{pmatrix} 2 & -1 \\ -1 & 1 \end{pmatrix} \begin{pmatrix} 0 & 1 \\ 2 & 1 \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix} = \begin{pmatrix} 2 & -1 \\ -1 & 1 \end{pmatrix} \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} = \begin{pmatrix} -1 & 0 \\ 2 & 2 \end{pmatrix}$$

- b) Give an example of a pair of inverse matrices that are similar to each other.

A trivial example is $\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$ and $\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}^{-1} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$.

These are not only similar, but equal.

(The change of basis matrix can be taken to be the identity).

Another example is $\begin{pmatrix} 2 & 0 \\ 0 & 1/2 \end{pmatrix}$ and $\begin{pmatrix} 2 & 0 \\ 0 & 1/2 \end{pmatrix}^{-1} = \begin{pmatrix} 1/2 & 0 \\ 0 & 2 \end{pmatrix}$.

since $\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 2 & 0 \\ 0 & 1/2 \end{pmatrix} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 0 & 2 \\ 1/2 & 0 \end{pmatrix} = \begin{pmatrix} 1/2 & 0 \\ 0 & 2 \end{pmatrix}$,

with $Q = Q^{-1} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$

8. a) Compute the determinant of this matrix:

$$A = \begin{pmatrix} 0 & 1 & 0 \\ 2 & 1 & 3 \\ -1 & 2 & 2 \end{pmatrix}$$

Using cofactor expansion along row 1:

$$\det A = 1 \cdot 0 \cdot \det \begin{pmatrix} 1 & 3 \\ 2 & 2 \end{pmatrix} + (-1) \cdot 1 \cdot \det \begin{pmatrix} 2 & 3 \\ -1 & 2 \end{pmatrix} + 1 \cdot 0 \cdot \det \begin{pmatrix} 2 & 1 \\ -1 & 2 \end{pmatrix}$$

$$= -\det \begin{pmatrix} 2 & 3 \\ -1 & 2 \end{pmatrix} = -(4 + 3) = \boxed{-7}$$

b) Give two examples of a 3×3 matrix with determinant 14.

Since the matrix above has determinant -7 and $\det$ is multilinear, we have $\det \begin{pmatrix} 0 & 1 & 0 \\ -4 & 1 & 3 \\ 2 & 2 & 2 \end{pmatrix} = 14$

by multiplying the first column by -2.

Since the determinant of a diagonal matrix is the product of its diagonal entries, we have

$$\det \begin{pmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 7 \end{pmatrix} = 14.$$

9. a) Find the eigenvalues and eigenvectors of this matrix:

If λ is an eigenvalue,
then $\det(A - \lambda I) = 0$.

$$\begin{vmatrix} 2-\lambda & 1 \\ 1 & 1-\lambda \end{vmatrix} = (2-\lambda)(1-\lambda) - 1 = \lambda^2 - 3\lambda + 1$$

$$\lambda^2 - 3\lambda + 1 = 0 \Rightarrow \lambda = \frac{3 \pm \sqrt{9-4}}{2} = \frac{3 \pm \sqrt{5}}{2}$$

$$\lambda_1 = \frac{3+\sqrt{5}}{2}; \left(\begin{array}{cc|c} 2-\frac{3+\sqrt{5}}{2} & 1 & 0 \\ 1 & 1-\frac{3+\sqrt{5}}{2} & 0 \end{array} \right) \rightarrow \left(\begin{array}{cc|c} \frac{1-\sqrt{5}}{2} & 1 & 0 \\ 1 & -\frac{1-\sqrt{5}}{2} & 0 \end{array} \right) \xrightarrow{R1 \leftrightarrow R2}$$

b) Is v an eigenvector of A ? Explain why or why not.

$$v = \begin{pmatrix} 1 \\ 1 \end{pmatrix} \quad A = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}$$

Since

$$Av = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 2 \\ 2 \end{pmatrix} = 2v$$

v is an eigenvector of A
with eigenvalue 2.

$$\left(\begin{array}{cc|c} 1 & -\frac{1-\sqrt{5}}{2} & 0 \\ \frac{1-\sqrt{5}}{2} & 1 & 0 \end{array} \right)$$

$$\xrightarrow{R2 - \frac{1-\sqrt{5}}{2} R1} \left(\begin{array}{cc|c} 1 & -\frac{1-\sqrt{5}}{2} & 0 \\ 0 & 0 & 0 \end{array} \right)$$

$$v_1 = k \begin{pmatrix} \frac{1+\sqrt{5}}{2} \\ 1 \end{pmatrix}$$

By the same process:

$$\lambda_2 = \frac{3-\sqrt{5}}{2}, \quad v_2 = k \begin{pmatrix} \frac{1-\sqrt{5}}{2} \\ 1 \end{pmatrix}$$

Name: Key

10. a) Compute an orthonormal basis for the subspace spanned by this set of vectors in $\mathbb{R}^3$:

$$\left\{ \begin{pmatrix} -1 \\ 0 \\ 3 \end{pmatrix}, \begin{pmatrix} 2 \\ 3 \\ 0 \end{pmatrix} \right\}$$

$$\begin{pmatrix} -1 \\ 0 \\ 3 \end{pmatrix} \cdot \begin{pmatrix} 2 \\ 3 \\ 0 \end{pmatrix} = -2 + 0 + 0 = -2, \quad \begin{pmatrix} -1 \\ 0 \\ 3 \end{pmatrix} \cdot \begin{pmatrix} -1 \\ 0 \\ 3 \end{pmatrix} = 1 + 0 + 9 = 10$$

$$\begin{pmatrix} 2 \\ 3 \\ 0 \end{pmatrix} - \frac{-2}{10} \begin{pmatrix} -1 \\ 0 \\ 3 \end{pmatrix} = \begin{pmatrix} 2 \\ 3 \\ 0 \end{pmatrix} + \frac{1}{5} \begin{pmatrix} -1 \\ 0 \\ 3 \end{pmatrix} = \begin{pmatrix} 9/5 \\ 3 \\ 3/5 \end{pmatrix}$$

So $\left\{ \begin{pmatrix} -1 \\ 0 \\ 3 \end{pmatrix}, \begin{pmatrix} 9/5 \\ 3 \\ 3/5 \end{pmatrix} \right\}$ is an orthogonal basis for the subspace.

rescaling $\rightarrow \left\{ \begin{pmatrix} -1 \\ 0 \\ 3 \end{pmatrix}, \begin{pmatrix} 9/5 \\ 3 \\ 3/5 \end{pmatrix} \right\} \rightarrow \left\{ \begin{pmatrix} -1 \\ 0 \\ 3 \end{pmatrix}, \begin{pmatrix} 3 \\ 5 \\ 1 \end{pmatrix} \right\} \rightarrow \boxed{\left\{ \begin{pmatrix} -1/\sqrt{10} \\ 0 \\ 3/\sqrt{10} \end{pmatrix}, \begin{pmatrix} 3/\sqrt{35} \\ 5/\sqrt{35} \\ 1/\sqrt{35} \end{pmatrix} \right\}}$ orthonormal basis

b) Give a basis for the subspace of $\mathbb{R}^4$ that is orthogonal to v .

$$v = \begin{pmatrix} 2 \\ 1 \\ 3 \\ -4 \end{pmatrix}$$

norm = $\sqrt{10}$ norm = $\sqrt{35}$

$$\left\{ \frac{1}{\sqrt{10}} \begin{pmatrix} -1 \\ 0 \\ 3 \end{pmatrix}, \frac{1}{\sqrt{35}} \begin{pmatrix} 3 \\ 5 \\ 1 \end{pmatrix} \right\}$$

Set $(2 \ 1 \ 3 \ -4)$ to be the row space of

a matrix. Then the nullspace of the matrix is the orthogonal complement to $\text{span} \left\{ \begin{pmatrix} 2 \\ 1 \\ 3 \\ -4 \end{pmatrix} \right\} \subseteq \mathbb{R}^4$

$$(2 \ 1 \ 3 \ -4) \xrightarrow{1/2 R_1} \begin{matrix} x & y & z & w \\ \begin{pmatrix} 1 & 1/2 & 3/2 & -2 \end{pmatrix} \end{matrix}$$

$$\text{So } \begin{pmatrix} x \\ y \\ z \\ w \end{pmatrix} = y \begin{pmatrix} -1/2 \\ 1 \\ 0 \\ 0 \end{pmatrix} + z \begin{pmatrix} -3/2 \\ 0 \\ 1 \\ 0 \end{pmatrix} + w \begin{pmatrix} 2 \\ 0 \\ 0 \\ 1 \end{pmatrix}, \quad \text{So } \left\{ \begin{pmatrix} -1/2 \\ 1 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} -3/2 \\ 0 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 2 \\ 0 \\ 0 \\ 1 \end{pmatrix} \right\}$$

is a basis for $(\text{span}\{v\})^\perp$.